

```
import pandas as pd

# Load the dataset
df = pd.read_csv('/content/extracted_nba_players_data.csv.csv')

# Display the first 5 rows to confirm loading and identify target variable
display(df.head())
```

	fg	3p	ft	reb	ast	stl	blk	tov	target_5yrs	total_points	efficiency
0	34.7	25.0	69.9	4.1	1.9	0.4	0.4	1.3	0	266.4	0.270073
1	29.6	23.5	76.5	2.4	3.7	1.1	0.5	1.6	0	252.0	0.267658
2	42.2	24.4	67.0	2.2	1.0	0.5	0.3	1.0	0	384.8	0.339869
3	42.6	22.6	68.9	1.9	0.8	0.6	0.1	1.0	1	330.6	0.491379
4	52.4	0.0	67.4	2.5	0.3	0.3	0.4	0.8	1	216.0	0.391304

```
from sklearn.model_selection import train_test_split

# Separate features (X) and target variable (y)
X = df.drop('target_5yrs', axis=1)
y = df['target_5yrs']

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Display the shapes of the resulting datasets
print(f"X_train shape: {X_train.shape}")
print(f"X_test shape: {X_test.shape}")
print(f"y_train shape: {y_train.shape}")
print(f"y_test shape: {y_test.shape}")
```

```
X_train shape: (1072, 10)
X_test shape: (268, 10)
y_train shape: (1072,)
y_test shape: (268,)
```

```
from sklearn.naive_bayes import GaussianNB

# Initialize the Gaussian Naive Bayes model
gnb = GaussianNB()

# Train the model using the training data
gnb.fit(X_train, y_train)

# Make predictions on the test set
y_pred = gnb.predict(X_test)

print("Model training and prediction complete.")
```

Model training and prediction complete.

```
from sklearn.metrics import confusion_matrix, precision_score, recall_score
import matplotlib.pyplot as plt
import seaborn as sns

# Generate the Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

# Calculate Precision
precision = precision_score(y_test, y_pred)

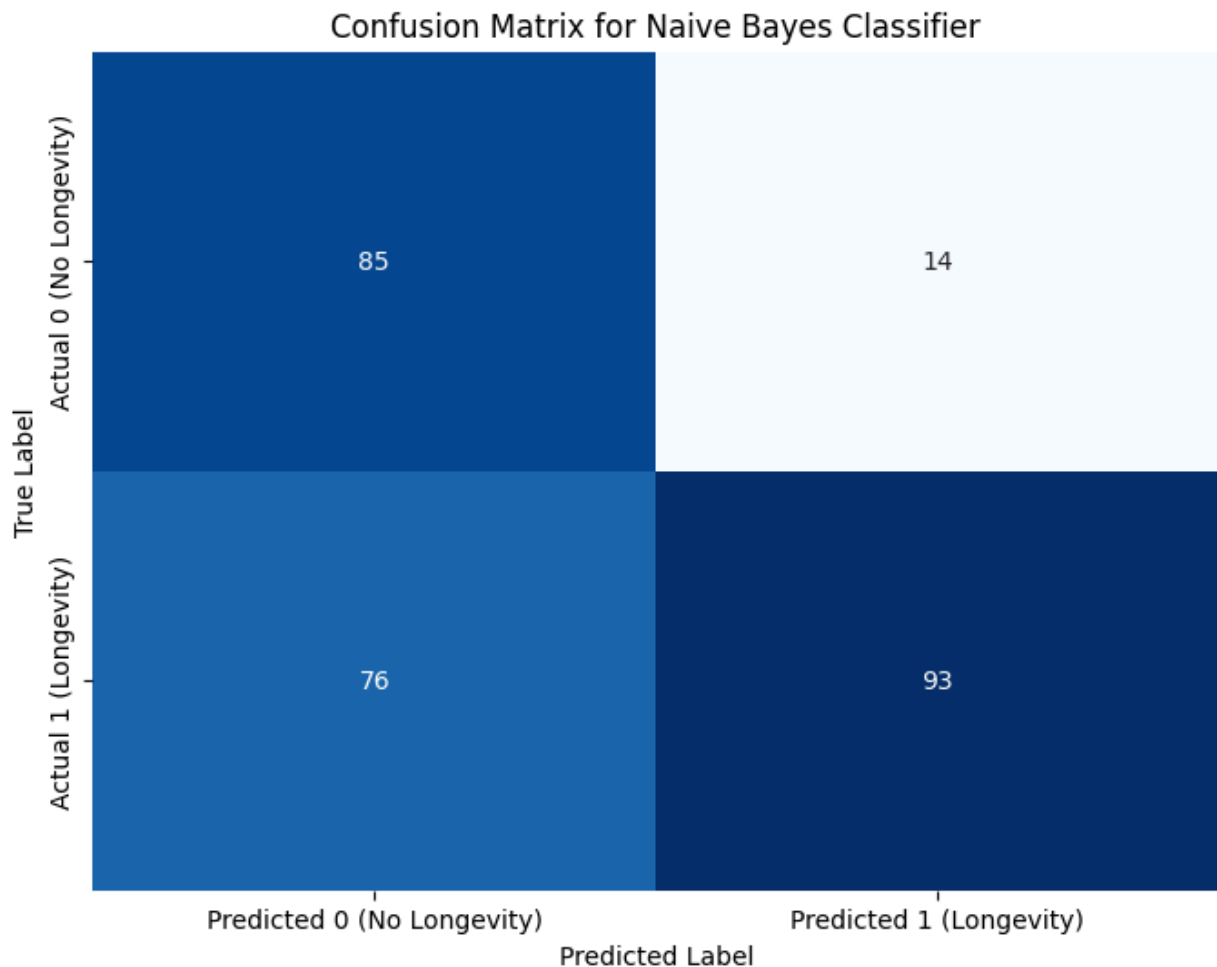
# Calculate Recall
recall = recall_score(y_test, y_pred)

print(f"Precision: {precision:.4f}")
print(f"Recall: {recall:.4f}")

# Visualize the Confusion Matrix
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', cbar=False,
            xticklabels=['Predicted 0 (No Longevity)', 'Predicted 1 (Longevity)'],
            yticklabels=['Actual 0 (No Longevity)', 'Actual 1 (Longevity)'])
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix for Naive Bayes Classifier')
plt.show()
```

Precision: 0.8692

Recall: 0.5503



Realism for Basketball Stats:

This assumption is generally unrealistic for basketball statistics. Consider the example of `total_points` and `efficiency`. These two features are inherently linked: a player who scores more points (contributing to `total_points`) is likely to have a higher efficiency rating. Similarly:

Points and Minutes Played: While not directly in our dataset, it's a classic example. More minutes on the court generally lead to more opportunities to score points. They are not independent. **Assists and Field Goal Percentage:** A player with a high assist count often plays a primary ball-handling role, which might influence their own shot attempts and thus their field goal percentage.

Rebounds and Blocks: Players good at rebounding (especially defensive rebounds) might also be strong defensive players, leading to more block opportunities. Because basketball statistics are highly interconnected and often correlate, the independence assumption is violated. However, despite this violation, Naive Bayes models can still perform surprisingly well in practice. This is often attributed to the fact that while the probabilities calculated by Naive

Bayes might be inaccurate, the ranking of probabilities (which is used for classification) can still be robust.

Model Reliability and Limitations for a Scouting Department

Reliability:

High Precision (0.8692): When the model predicts a player will have a 5-year career, it's correct about 87% of the time. This means the model is quite reliable in identifying potential long-term players, reducing the risk of drafting "busts" based on its positive predictions. This is valuable for making confident selections. **Simplicity and Speed:** Gaussian Naive Bayes is computationally efficient and fast to train, making it suitable for quick initial assessments.

Limitations:

Moderate Recall (0.5503): The model only identifies about 55% of the players who actually achieve a 5-year career. This means it has a significant number of false negatives (76 players), indicating that it will miss a substantial amount of genuine talent. If the scouting department relies solely on this model, they might overlook many future successful players. **Independence Assumption Violation:** As discussed, the model's core assumption is violated by the correlated nature of basketball statistics. While it can still perform adequately, it means the model's probabilistic outputs should be treated with caution, and its decisions might not perfectly reflect the underlying statistical relationships.

Lack of Feature Importance (Directly): Unlike some other models, Naive Bayes doesn't directly provide a clear ranking of feature importance in the same way. While we can infer some things, it's not as straightforward. **Binary Output:** The model provides a binary prediction (longevity or not) and doesn't offer nuanced insights into why a player is predicted to succeed or fail beyond the statistical probabilities for each feature. For scouting, more detailed explanations are often desired. **Actionable Recommendations for Scouting:**

Use as a Filter, Not a Final Decision-Maker: Given the high precision, the model can serve as an excellent initial filter to confidently identify a pool of players likely to succeed. These players warrant further in-depth scouting.

Beware of Missed Talent: The lower recall means scouts should be aware that the model will miss some valuable players. It should not be used to unilaterally dismiss players who receive a "no longevity" prediction without further human evaluation. Complement with Domain Expertise: Always integrate the model's predictions with experienced scouts' qualitative assessments, game film analysis, and player interviews. This model is a tool to assist, not replace, human expertise.